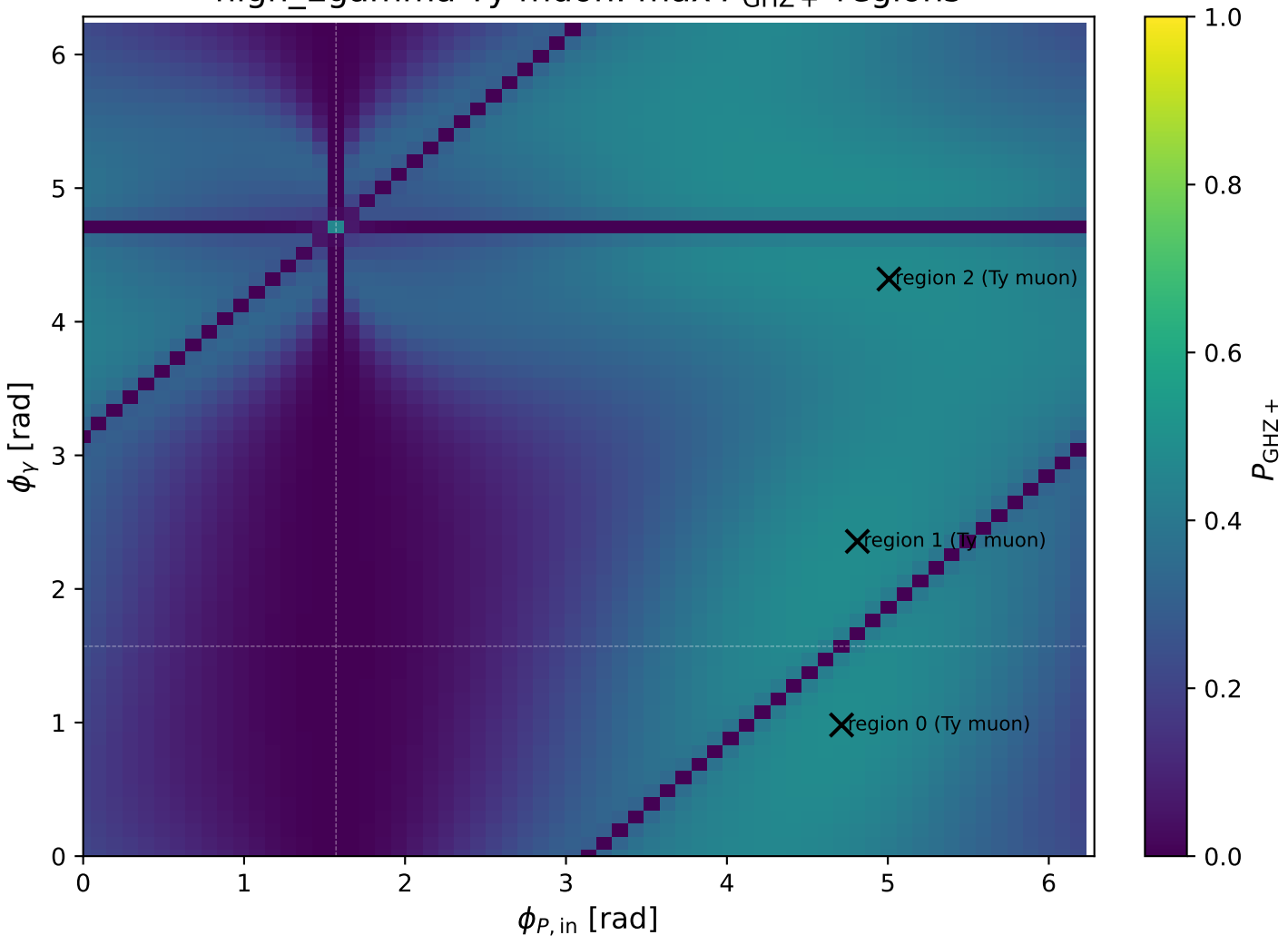
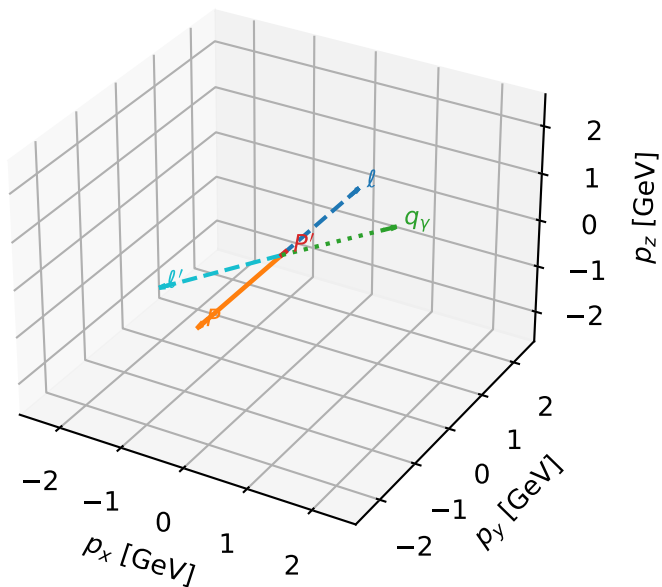


high_Egamma Ty muon: max $P_{\text{GHz}+}$ regions



Ty muon characteristic max P_{GHZ} + configuration

3D momenta



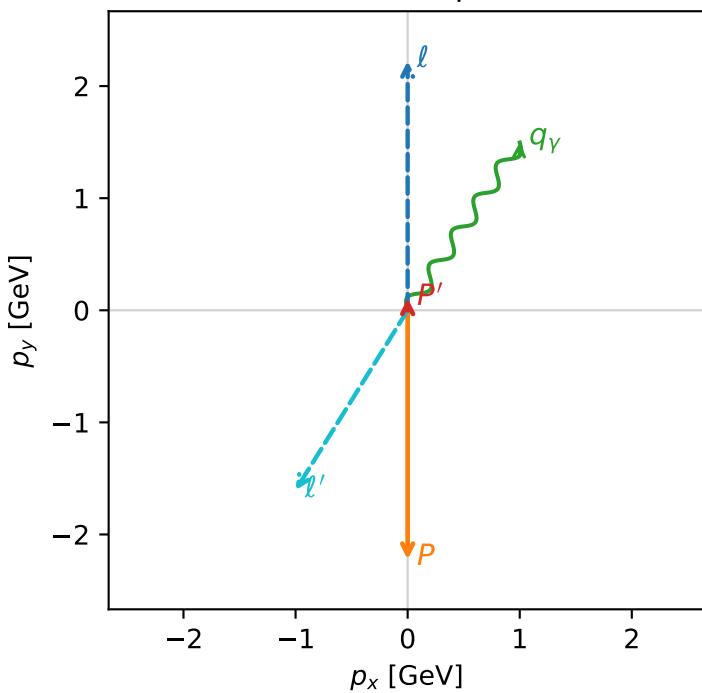
ghz_purity_high_egamma_ty_lepton_region_0 $P_{\text{GHZ}}=0.482446$
 region: high_Egamma_Ty_lepton_region_0
 outgoing-state purity: 0.50004
 kinematic point: high_s_high_theta_in_high_Egamma
 incoming state: $\frac{1}{2}\sum_{h_p}\rho(T_{y,\mu},h_p)$

$s=21.5158$, $\sqrt{s}=4.63852$, $|\vec{P}|=2.22311$, $|\vec{P}'|=0.108644$
 $\theta_{\text{in}}=1.57079$, $\phi_\ell=1.5708$, $\phi_P=4.71239$
 $E_\gamma=1.8$, $\phi_\gamma=0.981748$
 $Q^2=15.5469$, $x_B=0.834909$, $t=-3.28022$, $W^2=3.95402$, $y=0.902847$
 $\theta(\ell',\gamma)=178.171$ deg

four-momenta $[E, p_x, p_y, p_z]$ GeV:

ℓ	$E=$	2.2256	$p_x=$	0.0000	$p_y=$	2.2231	$p_z=$	-0.0000
P	$E=$	2.4129	$p_x=$	-0.0000	$p_y=$	-2.2231	$p_z=$	0.0000
ℓ'	$E=$	1.8942	$p_x=$	-1.0000	$p_y=$	-1.6053	$p_z=$	0.0000
P'	$E=$	0.9443	$p_x=$	0.0000	$p_y=$	0.1086	$p_z=$	0.0000
q_γ	$E=$	1.8000	$p_x=$	1.0000	$p_y=$	1.4966	$p_z=$	0.0000

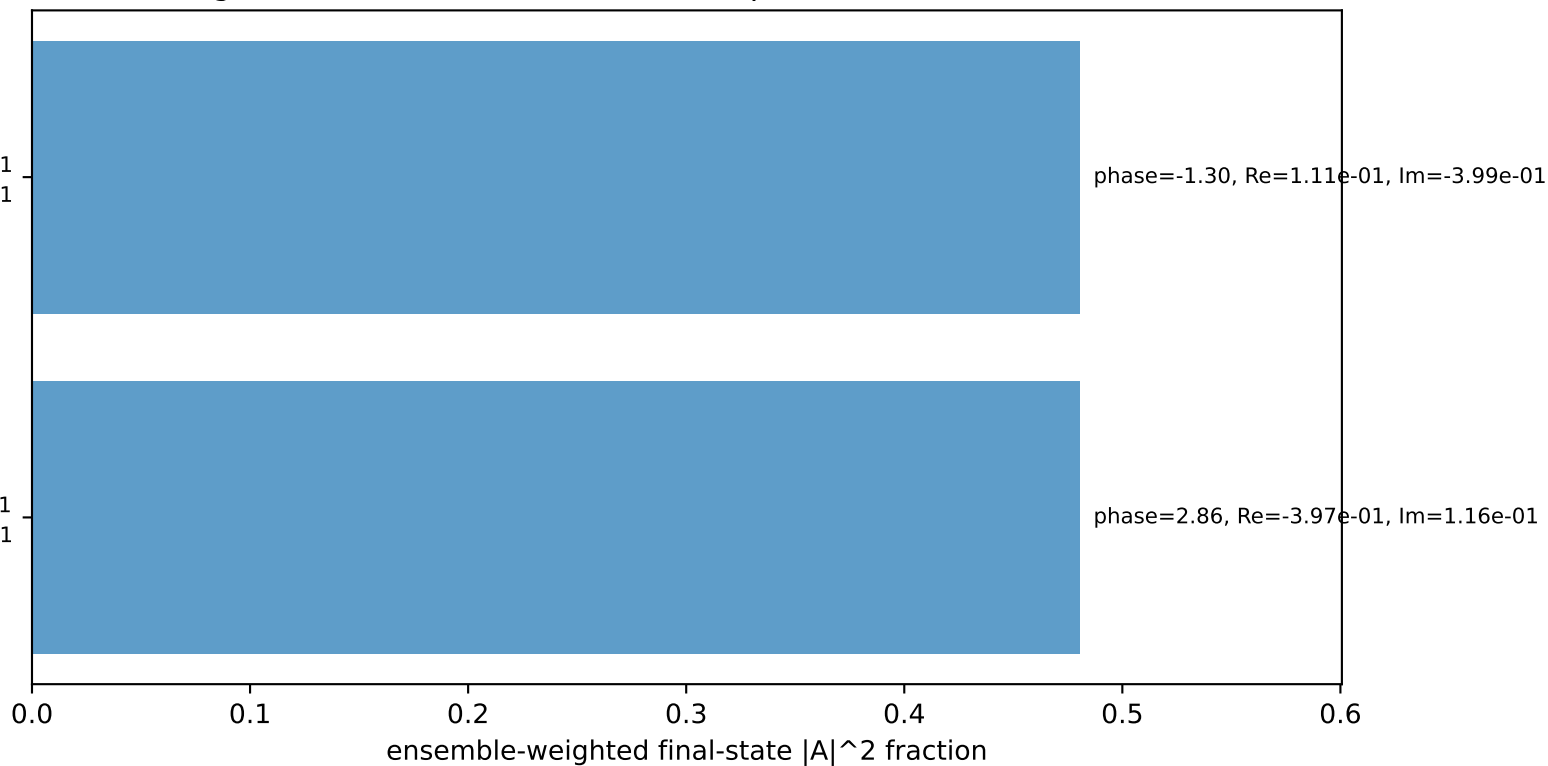
Transverse plane



$h_e=+1, h_p=+1, h_\gamma=+1$
 electron Ty, proton $h=+1$

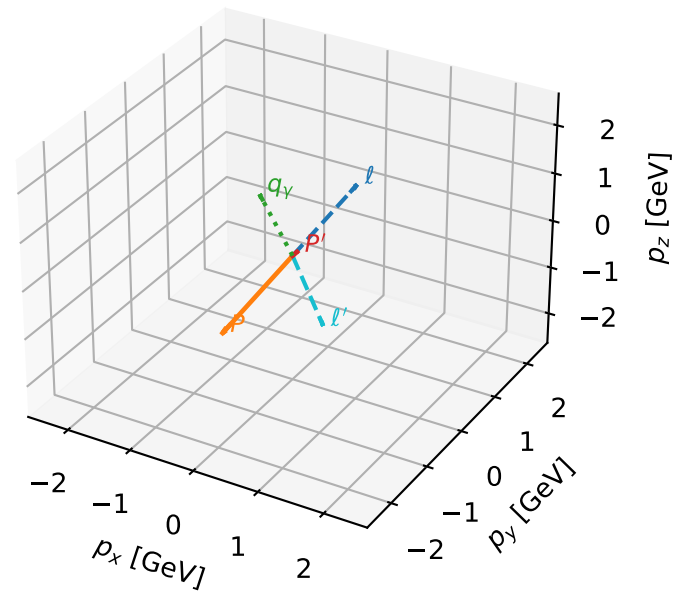
$h_e=-1, h_p=-1, h_\gamma=-1$
 electron Ty, proton $h=-1$

Leading initial-ensemble/final-state components (N=2, retained=96.1%)



Ty muon characteristic max $P_{\text{GHZ+}}$ configuration

3D momenta

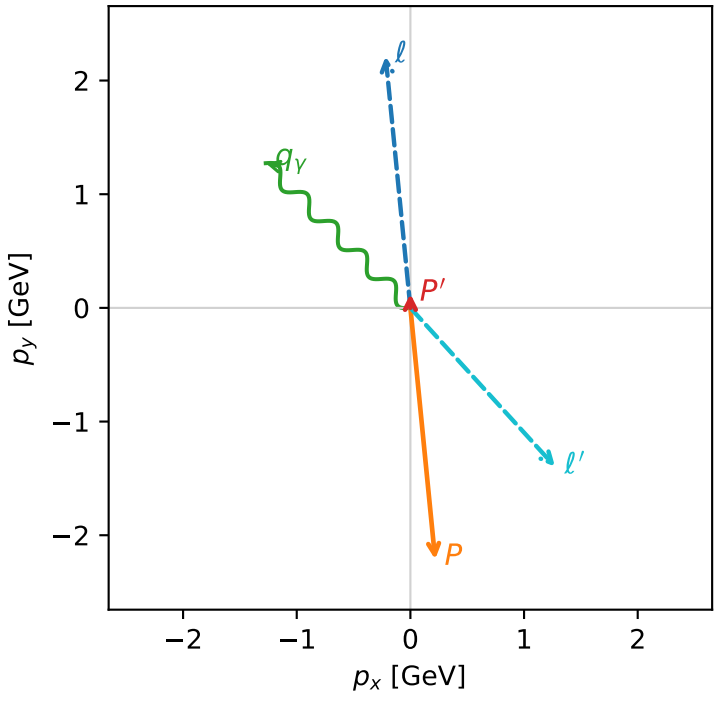


ghz_purity_high_egamma_ty_lepton_region_1 $P_{\text{GHZ+}}=0.480795$
region: high_Egamma_Ty_lepton_region_1
outgoing-state purity: 0.500036
kinematic point: high_s_high_theta_in_high_Egamma
incoming state: $\frac{1}{2}\sum_{h_p} \rho(T_{y,\mu}, h_p)$

$s=21.5158$, $\sqrt{s}=4.63852$, $|\vec{P}|=2.22311$, $|\vec{P}'|=0.123642$
 $\theta_{\text{in}}=1.57079$, $\phi_l=1.66897$, $\phi_p=4.81056$
 $E_V=1.8$, $\phi_V=2.35619$
 $Q^2=15.1349$, $x_B=0.830394$, $t=-3.35315$, $W^2=3.97111$, $y=0.883699$
 $\theta(l', \gamma)=177.348$ deg

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 2.2256$ $p_x= -0.2179$ $p_y= 2.2124$ $p_z= -0.0000$
 P $E= 2.4129$ $p_x= 0.2179$ $p_y= -2.2124$ $p_z= 0.0000$
 l' $E= 1.8924$ $p_x= 1.2728$ $p_y= -1.3964$ $p_z= 0.0000$
 P' $E= 0.9461$ $p_x= 0.0000$ $p_y= 0.1236$ $p_z= 0.0000$
 q_γ $E= 1.8000$ $p_x= -1.2728$ $p_y= 1.2728$ $p_z= 0.0000$

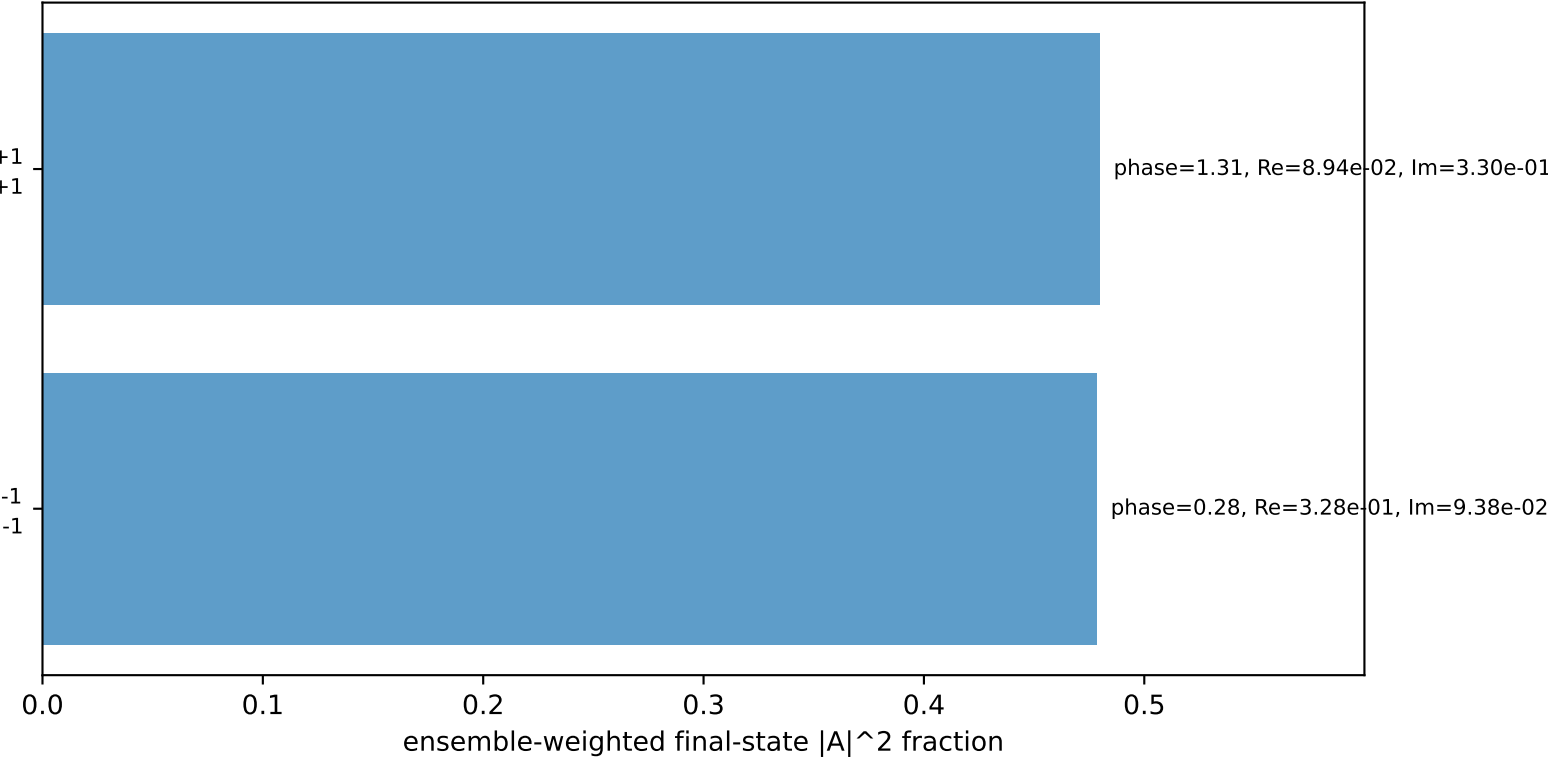
Transverse plane



$h_e=+1, h_p=+1, h_\gamma=+1$
electron Ty, proton h=+1

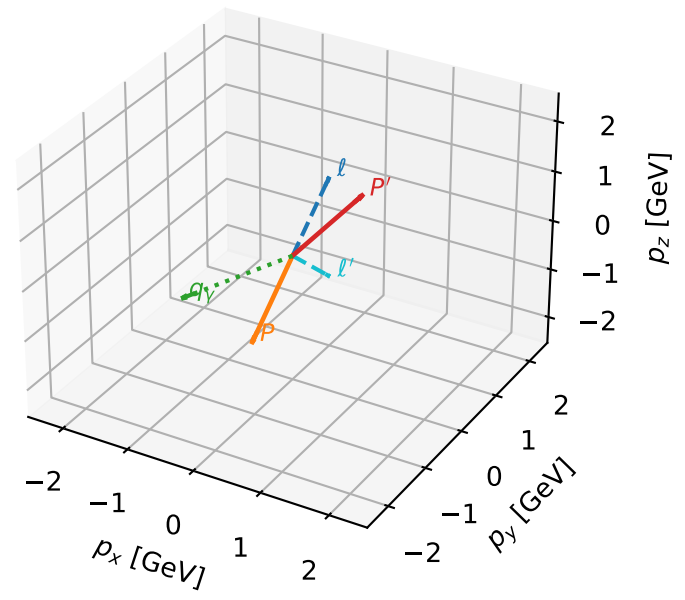
$h_e=-1, h_p=-1, h_\gamma=-1$
electron Ty, proton h=-1

Leading initial-ensemble/final-state components (N=2, retained=95.9%)



Ty muon characteristic max $P_{\text{GHZ+}}$ configuration

3D momenta

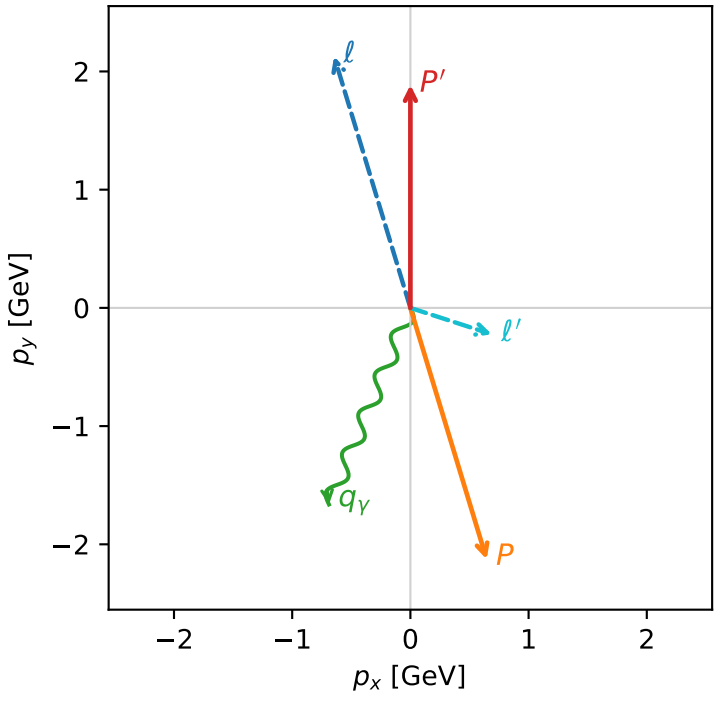


ghz_purity_high_egamma_ty_lepton_region_2 $P_{\text{GHZ+}}=0.478826$
region: high_Egamma_Ty_lepton_region_2
outgoing-state purity: 0.500008
kinematic point: high_s_high_theta_in_high_Egamma
incoming state: $\frac{1}{2}\sum_{h_p} \rho(T_{y,\mu}, h_p)$

$s=21.5158$, $\sqrt{s}=4.63852$, $|\vec{P}|=2.22311$, $|\vec{P}'|=1.88636$
 $\theta_{\text{in}}=1.57079$, $\phi_l=1.86532$, $\phi_p=5.00691$
 $E_V=1.8$, $\phi_V=4.31969$
 $Q^2=5.07463$, $x_B=0.268034$, $t=-16.4329$, $W^2=14.738$, $y=0.917958$
 $\theta(l', \gamma)=94.5328$ deg

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 2.2256$ $p_x= -0.6453$ $p_y= 2.1274$ $p_z= -0.0000$
 P $E= 2.4129$ $p_x= 0.6453$ $p_y= -2.1274$ $p_z= 0.0000$
 l' $E= 0.7318$ $p_x= 0.6888$ $p_y= -0.2234$ $p_z= 0.0000$
 P' $E= 2.1067$ $p_x= 0.0000$ $p_y= 1.8864$ $p_z= 0.0000$
 q_γ $E= 1.8000$ $p_x= -0.6888$ $p_y= -1.6630$ $p_z= 0.0000$

Transverse plane



$h_e=+1, h_p=+1, h_\gamma=+1$
electron Ty, proton $h=+1$

$h_e=-1, h_p=-1, h_\gamma=-1$
electron Ty, proton $h=-1$

Leading initial-ensemble/final-state components (N=2, retained=95.7%)

